

Data Visualizations: Association Rule Mining Results

This PDF contains the final visual evidence used to support the Apriori association rule mining report. Code snippets are included below each visualization to document how the charts were generated.

1. Support vs Confidence Scatter Plot

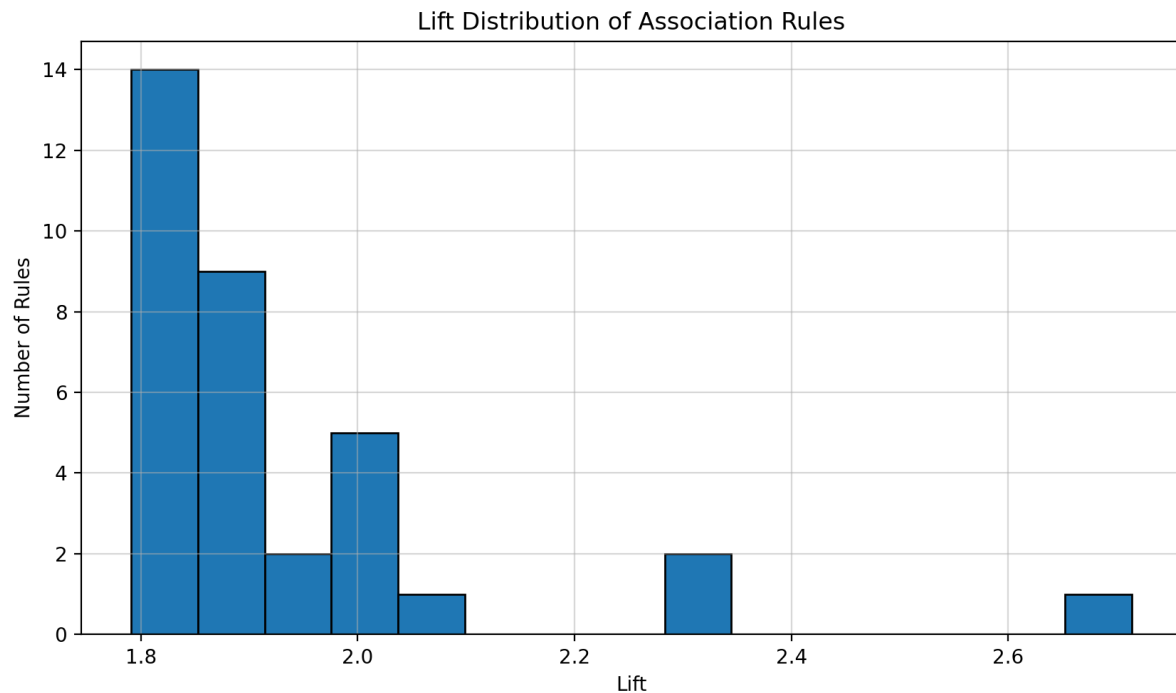


This scatter plot shows the reliability and frequency of the generated rules. Most rules have support between 0.05 and 0.08 and confidence above 0.60. Marker size represents lift, so larger markers indicate stronger associations compared to random chance.

Code used:

```
plt.scatter(rules['support'], rules['confidence'], s=rules['lift'] * 100, alpha=0.65)
plt.xlabel('Support')
plt.ylabel('Confidence')
plt.title('Association Rules: Support vs Confidence')
plt.grid(True)
plt.show()
```

2. Lift Distribution

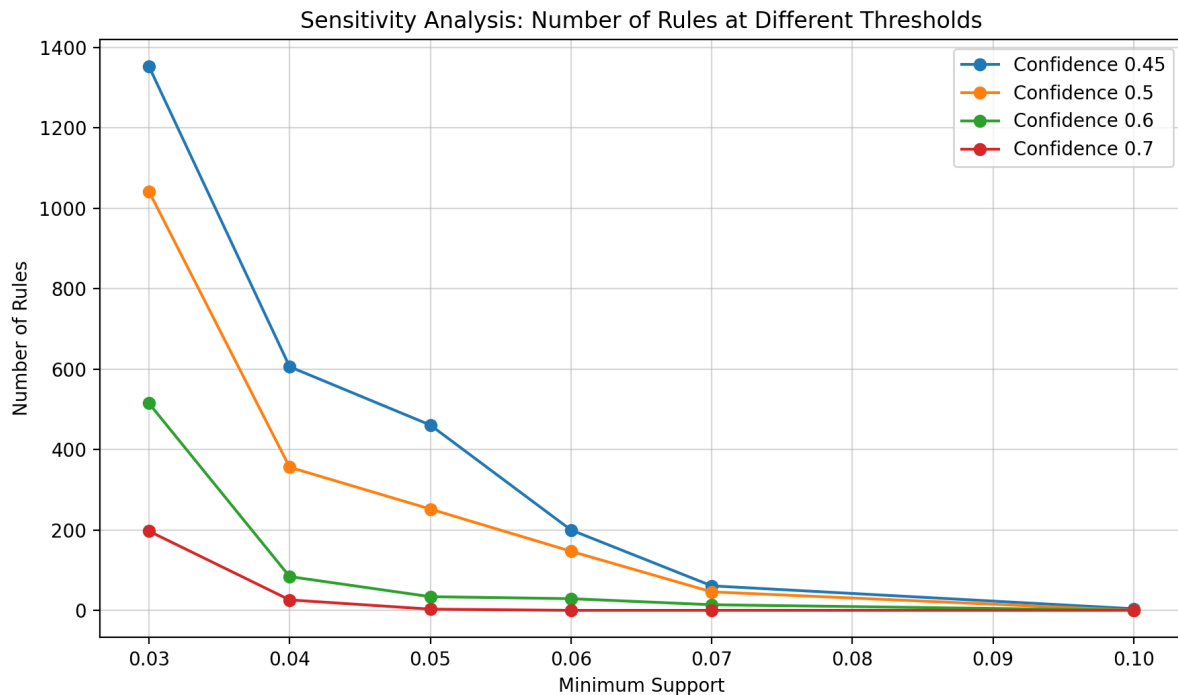


The lift distribution shows the strength of association rules. Most rules are concentrated around lift values from about 1.8 to 2.0, while a few rules exceed 2.0. Since all lift values are greater than 1.0, the selected rules represent positive associations.

Code used:

```
plt.hist(rules['lift'], bins=15, edgecolor='black')
plt.xlabel('Lift')
plt.ylabel('Number of Rules')
plt.title('Lift Distribution of Association Rules')
plt.grid(True)
plt.show()
```

3. Sensitivity Analysis

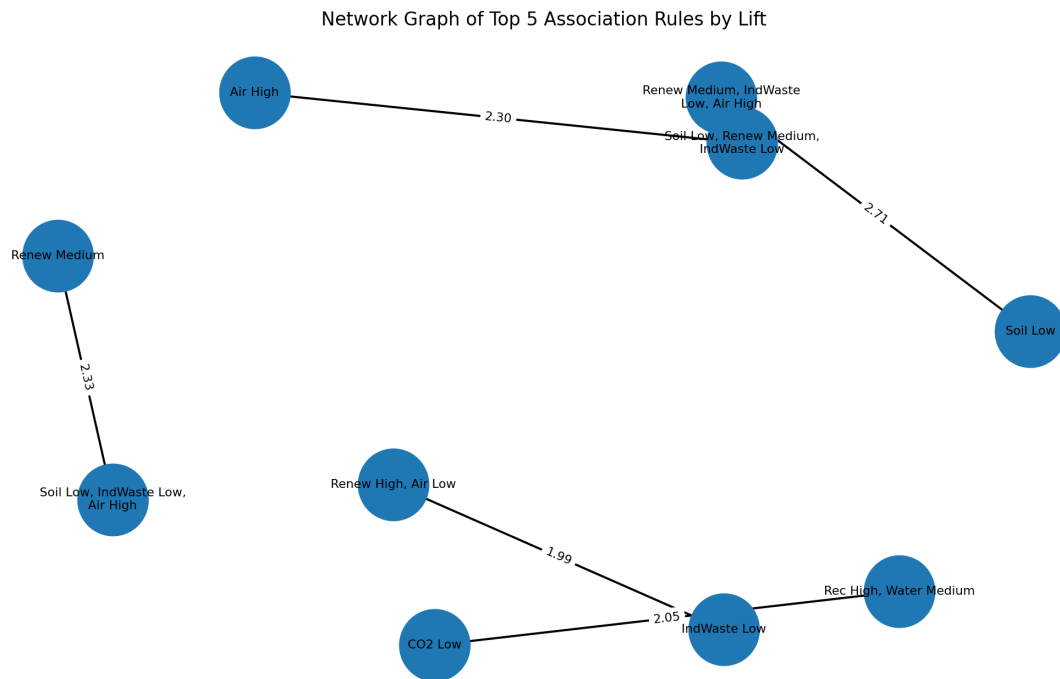


This chart shows how the number of generated rules changes when minimum support and confidence thresholds are adjusted. Lower thresholds create many more rules, but many may be less reliable. Higher thresholds produce fewer rules. The chosen support of 0.05 and confidence of 0.60 provides a balanced final rule set.

Code used:

```
for confidence in confidence_values:
    subset = sensitivity_df[sensitivity_df['min_confidence'] == confidence]
    plt.plot(subset['min_support'], subset['total_rules'], marker='o', label=f'Confidence {confidence}')
plt.xlabel('Minimum Support')
plt.ylabel('Number of Rules')
plt.title('Sensitivity Analysis: Number of Rules at Different Thresholds')
plt.legend()
plt.grid(True)
plt.show()
```

4. Network Graph of Top Rules by Lift

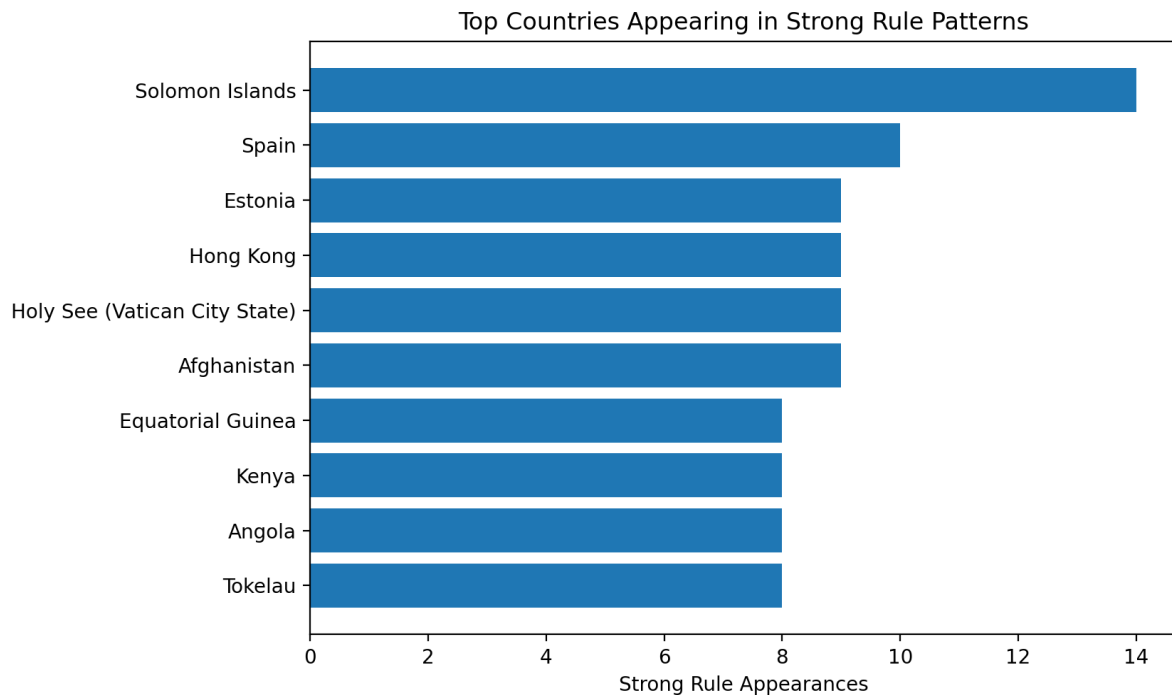


The network graph displays the top five rules by lift. Nodes represent itemsets and arrows represent association rules. Edge labels show lift values. This cleaner top-five version avoids overcrowding and highlights the strongest discovered relationships.

Code used:

```
top_rules_graph = strong_rules.sort_values(by='lift', ascending=False).head(5)
G = nx.DiGraph()
for _, row in top_rules_graph.iterrows():
    antecedent = ', '.join(list(row['antecedents']))
    consequent = ', '.join(list(row['consequents']))
    G.add_edge(antecedent, consequent, weight=row['lift'])
nx.draw_networkx(G, pos=nx.spring_layout(G, seed=42), arrows=True)
plt.show()
```

5. Country-Level Rule Pattern Frequency



This bar chart shows the countries that appear most frequently in rows matching the strong association rules. These countries are useful starting points for targeted pollution-control and energy-optimization recommendations.

Code used:

```
top_country = country_rule_df.head(10)
plt.barh(top_country['Country'][:-1], top_country['Strong_Rule_Appearances'][:-1])
plt.xlabel('Strong Rule Appearances')
plt.title('Top Countries Appearing in Strong Rule Patterns')
plt.show()
```

Visualization Summary

Together, these visualizations show that the selected rules are interpretable, statistically meaningful, and robust to threshold changes. The support-confidence plot confirms that the rules meet frequency and reliability thresholds. The lift distribution confirms that the rules are stronger than random chance. The sensitivity analysis justifies the selected support and confidence thresholds. The network graph communicates the strongest rule relationships, and the country chart supports targeted recommendations.